

UWB NLOS Identification and Mitigation based on Bidirectional Encoder Representations from Transformer (BERT) Deep Learning

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Abstract—The Non-Line-of-Sight (NLOS) phenomenon can hinder signal propagation and significantly reduce the accuracy of UWB for indoor positioning and navigation. The Channel Impulse Response (CIR) sequence generated during UWB ranging is widely used for channel identification. However, existing deep learning algorithms struggle to balance the local and global features of the CIR sequence effectively. To address this, this paper constructs a Bidirectional Encoder Representations from Transformers (BERT) channel identification model using the self-attention mechanism to improve the NLOS identification. The identification Accuracy, LOS recall, and F2 scores in multiple scenarios are 96.65%, 97.13%, and 0.9703 respectively. Comparing to state-of-art algorithms such as LS-SVM, CNN, and LSTM, our algorithm outperformed by 17.9%, 11.86%, and 10.80% respectively. For NLOS ranging errors, a fine-grained classification model is constructed with error correction model based on BERT. In multiple scenarios, the average NLOS ranging error is reduced by 41.97% and outperforms LS-SVM, CNN, and LSTM by 34.04%, 31.99%, and 16.81% respectively. In the overall positioning experiment, our algorithm achieves better performance than the existing algorithms by 32.13%.

Index Terms—Ultra-wideband (UWB), non-line-of sight (NLOS), deep Learning, Transformer

I. INTRODUCTION

With the advent of the artificial intelligence era, the demand for location information services has surged in technologies such as the Internet of Things [1], virtual reality, and human-computer interaction. Outdoor location services rely on mature GNSS [2] satellite navigation systems, which meet the needs of most application scenarios and have been successfully commercialized. However, there is currently no universal commercial solution for indoor positioning. Given that indoor activities account for a higher proportion of overall activities compared to outdoor activities, the potential for indoor positioning applications and market growth is significant.

Ultra-wideband (UWB) [3] is considered one of the most promising indoor positioning technologies due to its high precision time resolution, resistance to multipath interference, and low power consumption, making it highly suitable for complex indoor environments. However UWB ranging and positioning technology, which is based on time-of-flight (ToF) [4], faces interference from the Non-Line-of-Sight (NLOS) [5] phenomenon when the device's visibility path is obstructed. NLOS phenomenon can cause forward bias ranging errors, adversely affecting positioning accuracy. This phenomenon occurs frequently due to the complex and variable indoor layout, with numerous fixed and moving obstacles. Therefore, identifying and mitigating NLOS phenomena is crucial for improving UWB localization performance.

UWB NLOS identification involves two main components: channel features and classification algorithms. The sources of channel features include ranging information, location information, and channel impulse response (CIR). NLOS channel identification can be achieved by using a pre-trained threshold comparison classification algorithm, based on the variance of ranging results in different channel environments [6]. Although simple in principle, the accuracy of this algorithm is suboptimal. Location information can be obtained from either UWB location results [7] or other techniques [8], with the former requiring optimized UWB layout density and the latter introducing system complexity and additional constraints. The CIR sequence is recorded by hardware based on time information, capturing environmental noise, the direct signal from the LOS path, and multipath signal fluctuations from reflected paths. CIR features include manually extracted features and intelligent features derived from classification algorithms. Manually extracted CIR features are mainly statistical, including timing and energy characteristics, and are often

used with machine learning algorithms such as support vector machines (SVM) and decision trees. However, the recognition accuracy of a single manually extracted CIR feature combined with SVM [9] is only 60%. Combining different CIR features to form multi-dimensional feature vectors improve recognition accuracy. Deep learning algorithms such as convolutional neural networks (CNN) and long short-term memory (LSTM) can intelligently extract channel recognition features from CIR sequences, achieving accuracy higher than 80%. CNNs with CIR input sequences achieved an average NLOS identification accuracy of 90% across various scenarios. [10]. However, CNNs primarily focus on local features and struggle to capture channel characteristics in long CIR sequences. LSTM [11] is suited for processing time-series information, and increasing the number of hidden layers from one to two improved accuracy by 18.5%. However, LSTM's gradual data processing cannot avoid redundant information, leading to slow training speeds and gradient anomalies. In general, existing deep learning algorithms using CIR sequences face problems in the flow of information, such as over-emphasis on local features, inefficient boost processing, and gradient descent issue.

The NLOS identification algorithm labels ranging results as either LOS or NLOS. The ranging results of high-precision LOS-labeled can be directly used for positioning. In contrast, the ranging results of NLOS-labeled, which exhibit larger errors, must be mitigated before utilized for positioning. The main reason for ranging error is that obstacles interfere with true first path (FP_T) signal propagation, causing UWB's reported first path signal (FP_R) to be delayed and produce a positive bias error. Therefore, the core of range error mitigation is identifying the FP_T . The CIR sequence contains the FP_T , making it useful for range error mitigation. By combining manually extracted timing and energy CIR features, machine learning algorithms can map the CIR feature vector to the ranging error. For example, using SVM inputs from ranging results and CIR feature vectors (maximum CIR, average excess delay) reduced the mean UWB ranging error by 59.8% and the root mean square error by 49.3% in [12]. Range error mitigation algorithms improve accuracy by addressing errors but their effectiveness is limited to NLOS data. Thus their performance is highly dependent on the accuracy of NLOS identification algorithm. Additionally, existing range error mitigation algorithms struggle to combine CIR energy feature. This paper address aforementioned problems with a deep learning model based on Bidirectional Encoder Representations from Transformers (BERT) for channel identification and ranging error mitigation. BERT automatically extract key features from CIR, and its multi-head self-attention mechanism simultaneously focuses on multiple features to improve classification accuracy. BERT's bidirectional learning capability captures long-term dependencies within and between sequences. The original CIR sequence is first combined with BERT for NLOS identification, with optimized model and input parameters. Based on BERT identification results, we further establish another fine-grained range error classification system with a BERT model to perform range error mitigation.

II. BACKGROUND

A. UWB ToF Ranging and Location

UWB signal is emitted from the transmitter and traverse through various paths to reach the receiver and recorded as CIR. It represents the sum of multiple attenuated and delayed copies of the UWB pulse and expressed as [11]:

$$r(t) = \sum_{j=1}^J \alpha_j p(t - \tau_j) + \eta(t) \quad (1)$$

where α_j and τ_j are the amplitude and time delay of the j^{th} path, respectively; $p(\cdot)$ denotes the received UWB pulse; $\eta(t)$ is the noise. The UWB hardware employs the Leading Edge Detection (LDE) algorithm [13] to calculate the real-time environment noise level. When the CIR exceeds the dynamic threshold (LDE_{thre}), the corresponding time is considered the reported arrival time of the FP_R . The average flight time (t) of UWB signals between devices is calculated by measuring the time of FP_R to reach different devices, resulting in the corresponding ranging results. To reduce system errors and the impact of clock drift, this paper adopts the double-sided two-way ranging (DS-TWR) method, which records round-trips of signal propagation times. The ranging (d) formula for DS-TWR is given by $d = c * t$ where c is the speed of light.

In LOS conditions, the first path (FP) signal corresponding to the true ranging results and propagated through a direct visual path can be accurately captured. Theoretically, its signal arrival time (FP_T) is the same as FP_R . In NLOS conditions, the propagation of UWB signals can be interfered with by obstacles. Due to energy attenuation by obstacles, FP_T may not meet the dynamic LDE_{thre} , causing UWB to mistakenly use subsequent multipath signals as FP_R to calculate the ranging results. This forward delay (Δd) between FP_T and FP_R causes a forward bias error, affecting subsequent positioning accuracy. Therefore, it is crucial for UWB positioning to retain LOS ranging results and enhance the accuracy of NLOS ranging results through effective NLOS identification.

B. Bidirectional Encoder Representations from Transformers

As shown in Fig. 1, the input data for BERT [14] is encoded using WordPiece, Position and Segment Embedding. BERT inserts a special symbol 'CLS' before each independent input and uses the special symbol 'SEP' to separate different sequences within the same input. WordPiece Embedding assigns a unique integer to each input based on the corpus, expanding each integer into a vector of dimensions ($d_{emd} = 768$). Segment Embedding provides information about the sequence to which each input belongs, with values in the same sequence being identical. Position Embedding allows the network to determine the position of each data point and the distance between different data points in the sequence. For input X comprising two independent sequences, $A(x_1, x_2, \dots, x_n)$ and $B(x'_1, x'_2, \dots, x'_m)$, its encoder input Y is constructed by summing the corresponding token, segment, and positional embeddings, resulting in a dimension of $L(n + m + 2) \times d_{emd}$. The encoder comprises C identical

stacks, each containing two sub-layers: a multi-head self-attention [15] mechanism (*MultiHead*) and a position-wise fully connected feed-forward network (*FFN*). The multi-head attention has $D = 12$ Scaled Dot-Product Attention units. Its inputs are Queries (Q), Keys (K), and Values (V) given as:

$$Q_d = Y * W_d^Q \quad Q_d \in \mathbb{R}^{L \times d_k} \quad (2a)$$

$$K_d = Y * W_d^K \quad K_d \in \mathbb{R}^{L \times d_k} \quad (2b)$$

$$V_d = Y * W_d^V \quad V_d \in \mathbb{R}^{L \times d_v} \quad (2c)$$

where $d = 1 \dots D$, and the projections are parameter matrices $W_d^Q \in \mathbb{R}^{d_{emd} \times d_k}$, $W_d^K \in \mathbb{R}^{d_{emd} \times d_k}$ and $W_d^V \in \mathbb{R}^{d_{emd} \times d_v}$, $d_k = d_v = d_{emd}/C = 64$. Scaled dot product attention is:

$$Attention_d(Q_d, K_d, V_d) = softmax(\frac{Q_d \times K_d^T}{\sqrt{d_k}}) V_d \quad (3)$$

The MatMul operation of Q_d and K_d forms an attention map matrix of size $L \times L$. The Mask (optional) operation eliminates the weight information from the upper right corner of the attention map matrix. This ensures that each piece of data is only aware of its own attention weight and that of the preceding data. The scaling operation divides the data by $\sqrt{d_k}$ to prevent subsequent SoftMax function from entering the region of minimal gradients. SoftMax operation normalizes the weights with their sum equals 1. The normalized attention weight matrix with Q_d and K_d is then combined with V_d to produce the final matrix $Attention_d \in \mathbb{R}^{L \times d_v}$. *MultiHead* which consists of 12 Attention units, is given as:

$$MultiHead_c(Q, K, V) = Concat(head_1, \dots, head_{12}) W_c^O \quad (4a)$$

$$\text{where } head_d = Attention_d(Q_d, K_d, V_d) \quad (4b)$$

where $W_c^O \in \mathbb{R}^{(D \times d_v) \times d_{emd}}$ is the parameter matrix of the c^{th} Encoder where $c = 1 \dots C$, and the $MultiHead_c \in \mathbb{R}^{L \times d_{emd}}$. The subsequent layer after Multi-Head Attention is a layer normalization (Add&&Norm) with residual connection. In the c^{th} Encoder, the output of first Add&&Norm is $LayerNorm_c^1 = Norm(Y + MultiHead_c(Q, K, V))$. This output is fetched to the next sub-layer of Encoder which is a fully connected feed-forward neural network (*FFN*) that consists of two linear transformations connected by ReLU activation. The next layer after FFN is the second Add&&Norm with the output $LayerNorm_c^2 = Norm(FFN_c + LayerNorm_c^1)$. After the input vector Y passes through C encoders, we obtain the transformed vector $Z \in \mathbb{R}^{L \times d_{emd}}$, which includes a special token Z_{CLS} . The Z_{CLS} token is pivotal, serving not only as a marker for the entire input data but also as a representative of the entire input's information. Consequently, the vector of the Z_{CLS} token $\in \mathbb{R}^{1 \times d_{emd}}$ encapsulates the feature information of the entire Z sequence, and NLOS identification can be achieved by combining the classification layer.

III. PROPOSED METHOD

Fig. 2 illustrates the proposed UWB positioning strategy. It includes two main parts: NLOS channel identification and ranging error classification and mitigation. UWB tag

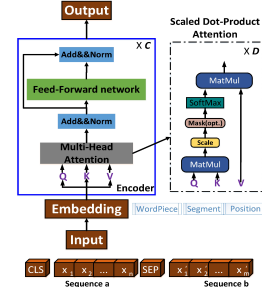


Fig. 1. Flowchart of BERT Framework.

obtains its ranging distance from multiple anchors and the corresponding CIR sequences. Based on the CIR sequence, a BERT-based NLOS channel identification model that classifies data as identified LOS(I-LOS) and identified NLOS (I-NLOS) is proposed. A fine-grained classification model for ranging errors is then designed to categorize the ranging results into high accuracy and low accuracy based on BERT channel identification result. High precision ranging results are directly used for positioning, while low precision ranging results are optimized before being used for positioning. Finally, based on the optimized ranging results, positioning coordinates are calculated using the least squares algorithm.

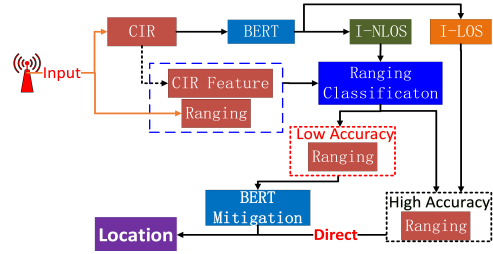


Fig. 2. Flowchart of NLOS identification and ranging mitigation.

A. NLOS Identification Based on BERT

NLOS channel identification separates high-precision LOS and NLOS ranging results to ensure optimal positioning performance. Besides FP_R , another key feature in the CIR sequence generated during distance measurement is the strongest path (SP), which generates the maximum CIR amplitude. In previous studies, we divided the CIR sequence into three stages: 1) Environment Noise Stage (ENS): CIR samples from the first index to the index of FP_R ; 2) CIR Steep Rise Stage (SRS): CIR samples from the index of $FP_R + 1$ to the index of SP ; 3) CIR Slow Descent Stage (SDS): CIR samples from the index of $SP + 1$ to the end index. As shown in Fig. 3, the ENS stage of LOS includes only natural fluctuations in environmental noise. With the arrival of UWB signals, the CIR increases and rapidly reaches its highest value due to weak multipath effects. As multipath signals reach the receiving end through longer paths of reflection or refraction, the CIR slowly decreases. In the NLOS channel environment, UWB signals are attenuated and cannot be detected by the LDE

algorithm in a timely manner. FP_R does not correspond to FP_T , but rather to a subsequent multipath signal of FP_R . Therefore, the ENS stage of NLOS will have additional high-energy abnormal CIR from UWB signals. Obstacles in NLOS conditions exacerbate multipath effects, preventing the CIR from quickly reaching its maximum value during the SRS stage. Under NLOS conditions, pronounced multipath effects delay the peak of the SP , resulting in higher fluctuations in the CIR during the subsequent SDS compared to LOS situations. BERT's self-attention mechanism allows the model to focus on these different stage differences, while bidirectional encoding simultaneously considers local and global features. The multi-head attention mechanism can focus on multiple features simultaneously. Using BERT with raw CIR sequence values can achieve channel identification and avoid the limitations and low generalizability of manually extracted CIR features.

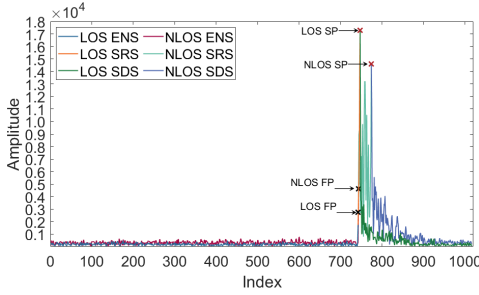


Fig. 3. CIR waveforms in different channel environments.

B. Ranging Error Classification and Mitigation

Positioning algorithms handle I-NLOS data by either deleting or correcting all I-NLOS data. Deleting I-NLOS data may cause high anchor density to ensure continuous positioning. Correcting I-NLOS data with a single model is generally ineffective. We propose a fine-grained classification and mitigation of I-NLOS ranging results with CIR features and BERT.

Vast majority of I-NLOS are true NLOS ranging results with high errors, but some of these NLOS ranging results can meet positioning requirements. Deleting them would result in loss of high-precision ranging results, while correcting them may lead to an increase in error. These I-NLOS primarily have abnormal CIR in the ENS and the attenuation of the FP_T energy by obstacles. The degree of UWB FP_T attenuation varies with physical characteristics of obstacles, leading to varying magnitudes of ranging errors. Therefore, this paper selects False Crests Number (FCN) and the difference between first path power and received power (e_{rise}) [16] as classification threshold features, reflecting the degree of first path signal attenuation. FCN [3] is a CIR threshold feature that indicates whether there is a high-energy abnormal CIR in the ENS stage. I-NLOS is proposed to classify into: NLOS with LOS ENS feature ($NLOS_{ENS}$), theoretical NLOS ($NLOS_{theory}$), and soft NLOS ($NLOS_{soft}$). We categorize NLOS data with $FCN = 0$ as $NLOS_{ENS}$ since CIR data has no energy anomalies in the ENS. It conforms to LOS waveform

characteristics, allowing it to be used for localization without correction. The remaining FCN values are greater than or equal to 1, indicating presence of energy anomaly CIR in ENS, and require full correction. We further classify these NLOS data with energy characteristic e_{rise} . It is known that when $e_{rise} \leq 6$, the impact of obstacles on UWB signals has a low effect on ranging accuracy [16]. NLOS data with $FCN \geq 1$ and $e_{rise} \leq 6$ will be classified as $NLOS_{soft}$ and data with $FCN \geq 1$ and $e_{rise} > 6$ as $NLOS_{theory}$.

We employ BERT algorithm for ranging errors that need correction. The output of the BERT is to produce Δd to identify the hidden FP_T in the CIR sequence corresponding to the true distance. However, the complete CIR sequence contains redundant data and limited energy information related to the real first path signal, which cannot provide evaluation data for the existing LDE algorithm on the FP_T . Traditional machine learning algorithms can only use CIR features, while deep learning algorithms can only use CIR sequences. Our BERT can focus on both CIR features and CIR sequences and fully exploit the hidden temporal and energy-related features to enhance the identification of the FP_T for distance mitigation. For $NLOS_{theory}$ with $FCN \geq 1$ and $e_{rise} > 6$, we propose to use First Path Error (FPE) [3] to provide time and related energy information about the FP_T . FPE represents the difference in time between the first path signal after optimizing the threshold and FP_R . Due to the presence of abnormal CIR data in ENS from multipath, root mean square delay extension (τ_{rms}) is also used to evaluate the multipath energy. For $NLOS_{soft}$, due to light obstacle attenuation, only FPE is sufficient as an input feature. In summary, we input CIR sequences and these features (FPE for $NLOS_{soft}$ and FPE with τ_{rms} for $NLOS_{theory}$) as BERT input vectors to obtain Δd . The mitigation distance $\hat{d} = d - \Delta d$.

IV. EXPERIMENT

To verify the performance of the proposed NLOS identification and ranging mitigation algorithms, the Decawave DW1000 UWB module system based on IEEE802.15.4-2011 has been used. Its effective communication distance is 50 m at communication rate of 110 kb/s. The transmission power is -41.3dBm/MHz at operating center frequency of 3.5GHz. Experimental campaign is at laboratory of University Ghent (UGent). As depicted in Fig. 4, the right side illustrates a typical office environment with office chairs, while the left side displays a more open area. In the open space, there are three sets of obstacles (green parallel stripes). Each set composes of physical walls, electronic devices (monitors, computer towers), and miscellaneous items, defined as mixed obstacles. In three sets of static experiments, mixed and human bodies were used as obstacles, and LOS and NLOS data were collected simultaneously during the movement process. Real positions are obtained from visual localization [17]. For each environment, the collected data is split into training (70) and testing (30) ratio.

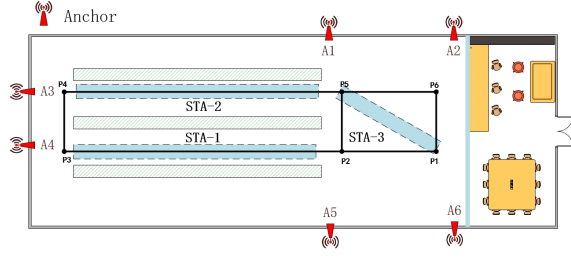


Fig. 4. Static and dynamic experimental equipment devices layout.

A. Performance of BERT NLOS identification

We evaluate the performance of the algorithm using three indicators: accuracy, recall of LOS ($Recall_{LOS}$), and F2 score.

$$Accuracy = \frac{LOS_T + NLOS_T}{LOS_T + LOS_F + NLOS_T + NLOS_F} \quad (5)$$

$$Recall_{LOS} = \frac{LOS_T}{LOS_T + LOS_F} \quad (6)$$

$$F2 = (1 + 2^2) * \frac{Accuracy * Recall_{LOS}}{(2^2 * Accuracy) + Recall_{LOS}} \quad (7)$$

where LOS_T and LOS_F refer to data correctly or falsely identified as the LOS, respectively; $NLOS_T$ and $NLOS_F$ refer to data correctly or falsely identified as the NLOS, respectively. The standard BERT includes 12 layers of encoders, but having too many encoders not only increases runtime but also affects accuracy. As the number of encoder layers increases, BERT's recognition accuracy improves because it can fully extract effective channel information from sequence data. However, with further increases in the number of encoder layers, the recognition accuracy decreases. This decline is due to overfitting and the merging of complex information, which results in the loss of some effective features. The optimal number of encoder layers is 5 with maximum runtime reduction of 51.77%. As shown in the Table I, compared with LS-SVM [12], CNN [10] and LSTM [11] with CIR sequences, our BERT outperform by 17.09%, 11.85%, and 10.80% respectively in terms of *Accuracy* with average as 96.65%. In terms of LOS recall($Recall_{LOS}$), our BERT outperforms by 16.06%, 17.18%, and 10.78% respectively with average recall as 97.13%. In terms of *F2*, our BERT also outperform the three existing algorithms by 19.73%, 19.64% and 11.95% respectively with average as 0.9703.

B. Performance of Ranging Mitigation

Our proposed ranging error mitigation algorithm exhibits superior performance than existing algorithms, resulting in a reduction of 41.97% in mean, 46.06% in STD, and 45.31% in RMSE. Fig.5 illustrates the improvement in ranging error over LS-SVM, CNN, and LSTM by 34.04%, 31.99%, and 16.81%.

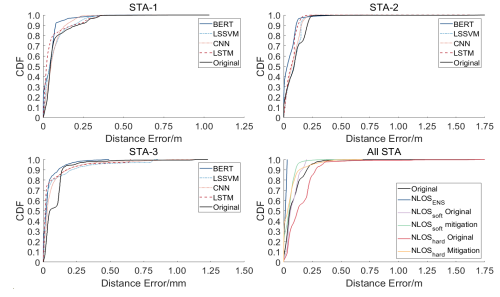


Fig. 5. The original and mitigation I-NLOS error distribution.

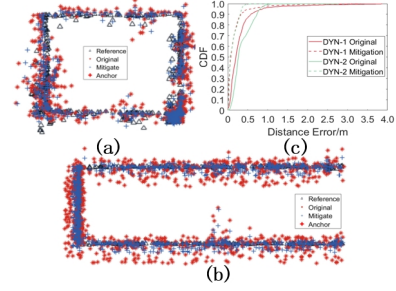


Fig. 6. Dynamic positioning experiment.

TABLE I
THE PERFORMANCE OF NLOS IDENTIFICATION

Scene	Method	Accuracy	$Recall_{LOS}$	<i>F2</i>
STA-1	BERT	96.46%	97.15%	0.9701
	LS-SVM	80.75%	81.09%	0.8102
	CNN	85.91%	84.93%	0.8512
	LSTM	87.59%	86.37%	0.8661
STA-2	BERT	96.65%	95.58%	0.9579
	LS-SVM	82.13%	80.09%	0.8049
	CNN	86.19%	85.63%	0.8574
	LSTM	89.61%	89.09%	0.8919
STA-3	BERT	96.84%	98.67%	0.9830
	LS-SVM	78.94%	81.22%	0.8075
	CNN	84.98%	85.49%	0.8539
	LSTM	86.04%	88.26%	0.8781

*The training parameters of LS-SVM are gam=8.7495 and sig2=0.4187 during identification and are gam=28.6462 and sig2=4.6439 during mitigation.

C. Positioning Experiment

To verify the improvement in positioning accuracy achieved by the BERT NLOS identification algorithm and ranging error mitigation algorithm, we conducted two dynamic experiment (DYN) with two types of obstacles (human body and mixed). As shown in Fig. 6(a), the motion trajectory consists of five consecutive instances of P1-P6-P5-P2-P1 in Fig. 4. The experimenter held the tag at chest level to ensure the presence of NLOS distance measurements. During the experiment, random movements of personnel within the scene will introduce interference to the UWB signals. As shown in Fig. 6, the black dot triangle represents the reference trajectory (RT) obtained through visual positioning while the red dot represents the original trajectory (OT) obtained from the raw UWB ranging

results without channel identification and mitigation, including NLOS ranging results. The blue cross represents the mitigated trajectory (MT) after BERT NLOS identification and ranging mitigation. Compared to the OT, the MT is closer to the RT. The proposed BERTs have successfully reduced the mean localization error from 0.3067m to 0.1623m with a significant performance improvement of 47.08%. Additionally, it reduces the STD localization error by 46.31% and RMSE localization error by 46.78%. The percentage of having less than 1 cm localization error increased from 32.22% to 57.37% with 25.15% improvement. As shown in Fig. 6(b), the motion trajectory consists of five distinct actions of P2-P3-P4-P5 in Fig. 4. The tag was mounted on a tripod and positioned higher than the experimenter, who moved the tripod to conduct the positioning experiment. Our algorithms reduces the mean positioning error to 0.1444m with a remarkable 59.12% reduction. Furthermore, the STD and RMSE of the positioning error were reduced by 59.82% and 51.49%, respectively. As shown in Fig. 6(c), The 90% (95%) positioning error has been reduced from 0.731m (0.802m) to 0.309m (0.349m), with reductions of 57.73% (56.48%). As shown in Table II, as compared to LS-SVM, CNN, and LSTM, our algorithms outperform in positioning accuracy, STD, and RMSE by 19.64%, 27.09%, and 30.33% respectively in DYN1. In DYN2, our algorithms outperform LS-SVM, CNN, and LSTM in positioning accuracy, STD, and RMSE by 15.91%, 26.48%, and 32.13% respectively.

TABLE II
POSITIONING PERFORMANCE IN DIFFERENT SCENARIOS

Scene		Mean(m)	Std(m)	Rmse(m)
DYN-1	Original	0.3067	0.4276	0.5101
	BERT	0.1623	0.2296	0.2715
	LS-SVM	0.2225	0.3455	0.4223
	CNN	0.2128	0.3388	0.4245
	LSTM	0.1998	0.3309	0.4148
DYN-2	Original	0.3532	0.5461	0.7975
	BERT	0.1444	0.2194	0.3869
	LS-SVM	0.2006	0.3640	0.6431
	CNN	0.1943	0.3366	0.6350
	LSTM	0.1902	0.3513	0.6323

V. CONCLUSION

This paper proposes a UWB NLOS channel identification and ranging error mitigation using BERT. In terms of *Accuracy* in multiple scenarios, our BERT algorithm outperforms existing algorithms such as LS-SVM, CNN, and LSTM by 17.90%, 11.86%, and 10.80% respectively. In terms of *Recall_{LOS}*, our BERT outperforms these existing by 16.06%, 17.18% and 10.78% respectively. Finally in terms of *F₂*, our BERT outperforms by 19.73%, 19.64% and 11.95% respectively. Based on the BERT NLOS channel recognition results, we further classify ranging errors into fine-grained categories using existing CIR features. We obtain the ranging error with a proposed BERT-based ranging error correction model with these fine-grained classified CIR sequences and CIR features. The proposed BERT algorithm corrected the I-NLOS ranging error with a reduction in mean, STD and

RMSE by 41.97%, 46.06%, and 45.31% respectively. The proposed BERT outperforms LS-SVM, CNN, and LSTM by 34.04%, 31.99%, and 16.81% respectively. Finally, in the dynamic positioning experiment campaign, the average positioning error, STD, and RMSE were improved by 53.10%, 53.06%, and 49.13%. The 90% and 95% positioning errors were also reduced by an average of 67.71% and 68.81%, respectively. Compared with existing algorithms, our algorithms demonstrates performance over LS-SVM, CNN and LSTM by 19.64%, 27.09%, and 32.13% respectively.

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